

PICOFLOW · CALCULATION NOTE

PicoFlow sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); 125 mm penstock, 33 mm inserts, results against PCF-REQ-001 v0.4
0.1	2026-09-25	Amish Chadha	First-principles sizing for TRL 3 against PCF-REQ-001 v0.3

PicoFlow sizing calculations

By calculation, the design meets 13 of the 17 requirements in PCF-REQ-001 v0.4. None is not met, one is at risk (R12, printed runner life) and three cannot be verified at TRL 3. Version 0.1 of this note found that 20 m of 110 mm PVC plus the fittings and the two nozzle branches lose about 24 % of the head at 10 L/s, which left the design point short of R3 (80 W) and R5 (40 %). On 2026-09-25 Amish accepted the recommendations in PCF-DDR-002: the design-point penstock is now 125 mm, R4 is 20 W at 1.0 m, R2 is stated per head, the R14 definition is confirmed, and the \$2 cost margin is accepted. With the 125 mm penstock the loss falls to 17 % of head and the design point gives 82.8 W into the battery (43.2 % water to wire). The two safety-critical items still pass on paper. A hardware clamp holds the DC bus below 48 V (R8), and a gate valve closing over 10 s or more keeps the drainage-grade penstock at about 33 kPa peak (R17).

Every number in this note is printed by `docs/04-calcs/sizing.py` (run `python docs/04-calcs/sizing.py` from the repo root), which also writes `docs/04-calcs/results.json` for the concept media and the drawing. The script reads geometry from `cad/src/model.py` and costs from `bom/bom.csv`.

Safety: PicoFlow works beside moving water, has a spinning runner and coupling, a generator that can reach about 80 V DC open circuit if both the load and the clamp fail, resistors that run hot, and a lithium battery. The numbers here are paper estimates, not evidence that a built unit is safe.

■ 1. Results against requirements

Table 1 lists every requirement, with the items at risk first. Status is one of met, not met, at risk, or not verifiable at TRL 3.

Table 1. Requirement status at TRL 3.

ID	Target	Value from this note	Status
R12	Bearing L10 20,000 h or more; runner 8,760 h or more	Bearing L10 about 8.7×10^6 h at the worst load; printed runner life unknown	At risk (runner)
R7	12 V charging, speed within 10 %, 100 % diversion within 1 s	300 W dump load is 1.68 times the largest output (178.8 W); control behavior needs a bench test	Not verifiable at TRL 3
R11	Runner or inserts in 30 min, bearings in 60 min; bearings above the spray	Bearings 392 mm above tailwater on the lid, outside the housing; times need a trial	Not verifiable at TRL 3
R16	Half the dry-season flow at most; all water returned	Depends on the site survey	Not verifiable at TRL 3
R1	Generates from 1.0 to 3.0 m with the same runner	DC bus 20.2 V at 1.0 m, above the 15 V the buck converter needs	Met
R2	5 to 10 L/s at 1.0 m; 5 to 15 L/s from 2.0 m	1.3 to 11.3 L/s at 1.0 m, 1.9 to 16.1 L/s at 2.0 m, 2.3 to 19.8 L/s at 3.0 m	Met
R3	80 W or more into the battery at 2.0 m, 10 L/s	82.8 W (84.3 W with inserts sized for exactly 10 L/s)	Met, 3.5 % margin
R4	20 W or more at 1.0 m, 7 L/s	26.3 W	Met
R5	40 % or more water to wire	43.2 %	Met
R6	1.8 kWh/day or more at the design point	1.99 kWh/day	Met
R8	No exposed conductor above 60 V DC; survives runaway	Clamp holds the bus at 39.6 V (below 48 V) in the worst case; 79.5 V open circuit only if the clamp also fails, inside the enclosed 100 V DC side; rim stress 0.08 MPa	Met by calculation; generator overspeed rating to confirm
R9	Prints in 220 x 220 x 250 mm, runner 24 h or less	Runner 200 x 200 x 50 mm, about 525 g, about 21 h	Met (estimate)
R10	Screen 6 mm or less, nozzles 20 mm or more	6 mm screen; inserts 20 to 45 mm; design insert 33 mm	Met by design
R13	Nozzle centerline 0.3 m or less above tailwater	280 mm in the model	Met
R14	Heaviest item 15 kg or less; turbine unit (items 1 to 6 and 11) 25 kg or less	Generator 7.0 kg; turbine unit 22.2 kg; manifold and valve 5.2 kg carried separately	Met (definition confirmed, PCF-DDR-002)
R15	Kit \$450 or less	\$448.00 with a new generator; \$378.00 with a salvaged one	Met; \$2 margin accepted by Amish (PCF-DDR-002), prices to be firmed with quotes before any build
R17		33.3 kPa at 3.0 m	Met

ID	Target	Value from this note	Status
	Peak penstock pressure 50 kPa or less with the valve closed over 10 s or more		

■ 2. Assumptions

- Water at 15 °C: density 1,000 kg/m³, kinematic viscosity 1.14×10^{-6} m²/s; $g = 9.81$ m/s².
- Design point (PCF-REQ-001): 2.0 m gross head from the forebay surface to the nozzle centerline, 10 L/s, 20 m of 125 mm drainage PVC (SN8, 3.7 mm wall, 117.6 mm bore), roughness 0.0015 mm. The 110 mm pipe of v0.1 (SN4, 3.2 mm wall, 103.6 mm bore) is kept in Table 5 for comparison.
- Penstock fittings, as loss coefficients on the pipe velocity head: rounded entrance 0.20, intake screen 0.10, two bends 0.30 in total, open gate valve 0.15. Friction factor from the Swamee-Jain equation. The gate valve is taken as full bore; see the note in section 11.
- Branches: 90 mm PVC, 84.0 mm bore, from a 125 x 90 mm reducing tee. Jet 1 runs around the housing (1.40 m, tee branch 1.0 plus four elbows at 0.3); jet 2 is direct (0.30 m, tee branch 1.0 plus one elbow). Lengths come from the model layout.
- Nozzle velocity coefficient 0.97, so the nozzle loses 6 % of the head it receives.
- Runner: bucket speed 0.46 of jet speed at best efficiency, pitch diameter 150 mm, jet-to-shaft efficiency 75 % for a first printed runner (laboratory runners reach 87 to 91 %, per Williamson, Stark and Booker, 2013, cited in PCF-PRB-001), runaway at 2.0 times best speed (upper end of 1.8 to 2.0).
- Generator: 10 rpm per volt of rectified open-circuit DC; iron and friction loss 12 W at 340 rpm, proportional to speed; winding and cable resistance 0.8 Ω (DC equivalent); rectifier drop 1.6 V. These are typical of a low-speed permanent magnet machine and must be measured on the chosen unit.
- Buck converter 94 %; battery 13.6 V while charging; the converter needs a bus of about 15 V to reach 14.4 V absorption.
- Bearings: 6204-2RS with a dynamic load rating of 13.5 kN (catalog class).
- Printed runner: PETG at 1,270 kg/m³, printed mass 0.6 times the solid massing volume (thin buckets, partial infill), 25 g/h on a 0.4 mm nozzle.
- Surge: PVC modulus 3.0 GPa, water bulk modulus 2.2 GPa; the drainage pipe and joints are assumed good for 50 kPa (0.5 bar, the common joint tightness test level for non-pressure pipe). This rating must be confirmed with the pipe supplier.
- Costs are indicative prices from `bom/bom.csv` ; the penstock (item 9) and battery (item 17) are excluded, as decided with the budget.

■ 3. Hydraulics and nozzle size

At the design point the nozzles need a 33.4 mm jet for exactly 10 L/s; the 33 mm insert gives 9.76 L/s (Table 2). The penstock carries water at 0.90 m/s and loses 0.159 m. The branches lose 0.099 m (jet 1, around the housing) and 0.055 m (jet 2), so the two jets carry 4.85 and 4.91 L/s, within 1.5 % of each other. The mean net head at the nozzles is 1.66 m, a loss of 17.0 % of the gross head (23.6 % with the 110 mm penstock of v0.1). With the 63 mm branches of the TRL 2 concept the loss would be 27.0 %, so the 90 mm branches stay.

Table 2. Design point hydraulics (2.0 m, 125 mm penstock, two 33 mm jets).

Quantity	Value
Flow	9.76 L/s
Penstock loss	0.159 m
Branch loss, jet 1 / jet 2	0.099 m / 0.055 m
Net head at the nozzles	1.66 m
Pipe and branch loss	17.0 % of gross head
Jet velocity	5.54 m/s
Jet to pitch diameter ratio	0.22

The jet to pitch diameter ratio of 0.22 is below the 0.28 of the PowerSpout DIY rotor cited in PCF-PRC-001, so 33 mm jets suit the 150 mm pitch circle.

■ 4. Power chain at the design point

Table 3. Power at each stage, 2.0 m and 10 L/s (9.76 L/s with the 33 mm inserts).

Stage	Power	Loss
Gross hydraulic power	191.6 W	
At the nozzles	159.0 W	Pipe and branches, 33 W
Jet	149.6 W	Nozzle, 9 W
Runner shaft	112.2 W at 324 rpm, 3.31 N·m	Runner, 37 W
Rectified DC	88.1 W at 28.3 V, 3.11 A	Generator and rectifier, 24 W (efficiency 78.5 %)
Into the battery	82.8 W, 6.09 A at 13.6 V	Buck converter, 5 W

Water to wire is 43.2 % and the daily energy, running 24 h, is 1.99 kWh. The generator runs at 32.4 V open circuit at best speed, so the bus is well above the battery.

■ 5. Head range

With the 33 mm inserts left in place, flow and output rise with head (Table 4). The bus stays above 15 V down to 1.0 m, so a 12 V battery charges across the whole range with a buck converter (R1). At 1.0 m and 7 L/s (R4) the right insert is again about 33 mm, and output is 26.3 W at 228 rpm with the bus at 20.1 V, above the 20 W target.

Table 4. Head range with the design-point inserts.

Gross head	Flow	Loss	Best speed	Open-circuit DC	Bus under load	Into battery	Water to wire	Runaway	Runaway DC
1.0 m	6.89 L/s	17.4 %	229 rpm	22.9 V	20.2 V	26.0 W	38.4 %	457 rpm	45.7 V
1.5 m	8.45 L/s	17.2 %	281 rpm	28.1 V	24.7 V	51.9 W	41.7 %	561 rpm	56.1 V
2.0 m	9.76 L/s	17.0 %	324 rpm	32.4 V	28.3 V	82.8 W	43.2 %	649 rpm	64.9 V
2.5 m	10.92 L/s	16.9 %	363 rpm	36.3 V	31.5 V	117.8 W	44.0 %	726 rpm	72.6 V
3.0 m	11.97 L/s	16.7 %	398 rpm	39.8 V	34.3 V	156.4 W	44.4 %	795 rpm	79.5 V

Flow range (R2)

From one 20 mm jet to two 45 mm jets, the inserts pass 1.3 to 11.3 L/s at 1.0 m, 1.9 to 16.1 L/s at 2.0 m and 2.3 to 19.8 L/s at 3.0 m. Two jets carrying 15 L/s would need 56.8 mm bores at 1.0 m, 42.8 mm at 2.0 m and 37.6 mm at 3.0 m. A 57 mm jet on a 150 mm pitch circle is far outside Turgo practice, which is why R2 is now stated as 5 to 10 L/s at 1.0 m and 5 to 15 L/s from 2.0 m (PCF-DDR-002). Both ranges are met.

Sensitivity

Pipe loss dominates the result. Table 5 resizes the inserts for 10 L/s in each case.

Table 5. Sensitivity at 2.0 m and 10 L/s.

Case	Loss	Into battery	Water to wire	Daily energy	At 1.0 m, 7 L/s
110 mm penstock (103.6 mm bore), runner 75 % (v0.1 baseline)	23.7 %	77.4 W	39.4 %	1.86 kWh	23.9 W
125 mm penstock (117.6 mm bore), runner 75 % (design)	17.5 %	84.3 W	43.0 %	2.02 kWh	26.3 W
160 mm penstock (150.6 mm bore), runner 75 %	12.2 %	90.1 W	45.9 %	2.16 kWh	28.4 W
110 mm penstock, runner 80 %	23.7 %	82.6 W	42.1 %	1.98 kWh	25.8 W
125 mm penstock, runner 80 %	17.5 %	89.9 W	45.8 %	2.16 kWh	28.4 W

(The design row differs from Table 3 by 1.5 W because its inserts are sized for exactly 10 L/s rather than rounded to 33 mm.)

With the 125 mm penstock, R3 has a margin of 2.8 W (3.5 %) with the rounded inserts. A runner below the assumed 75 % would erase it, so runner efficiency is the number to measure first at TRL 4.

■ 6. Runaway and the voltage clamp (R8)

If the battery is lost and the dump-load switch fails, the runner speeds up toward runaway. With the design inserts at 3.0 m that is 795 rpm and 79.5 V DC open circuit. The largest shaft power in the operating envelope is 241.5 W, at 3.0 m and 15 L/s (37.6 mm inserts); there the bus under MPPT load is 32.0 V and runaway is 768 rpm and 76.8 V.

The clamp is a comparator on the DC bus, independent of the microcontroller, that switches an 8.2 Ω resistor across the bus at 48 V and releases it at 40 V. The turbine torque is taken to fall linearly from stall to zero at runaway. With the resistor connected in the worst case, the runner settles at 451 rpm with the bus at 39.6 V and 192 W in the resistor. That is below the 40 V release point, so the clamp cycles between 40 and 48 V and the bus never exceeds 48 V. 8.2 Ω is still the largest E12 value that holds the equilibrium below 40 V, but the margin is now only 0.4 V; a 6.8 Ω resistor would give more margin if the generator constants come out less favorable. At 48 V the 8.2 Ω resistor dissipates 281 W, so it needs a 300 W aluminium-clad resistor on a heat sink. In normal running the bus never reaches the clamp: its highest open-circuit value at best speed is 39.8 V.

Only a double fault (load and clamp) lets the generator reach about 80 V, which is why the whole DC side stays enclosed and rated for 100 V. Mechanically, runaway is benign for the runner: the rim runs at 8.0 m/s and the hoop stress is about 0.08 MPa, a small fraction of PETG strength. The generator's overspeed rating at about 800 rpm must be confirmed with its supplier.

■ 7. Bearings (R12)

The worst bearing load is at 3.0 m and 15 L/s with one jet closed, so the remaining jet's force is not balanced by the other. The radial load is 42.4 N (jet force at the pitch radius). The axial load, taken conservatively as the full jet momentum plus runner weight, is 103.4 N. With $X = 0.56$ and $Y = 2.0$, the equivalent load is 230.6 N and the basic L10 life at 384 rpm is about 8.7×10^6 h, so bearing fatigue is not a concern. Life will be set by seal wear and water ingress, which a paper calculation cannot verify. The printed runner's life under continuous flow with sand is also unknown, so R12 remains at risk.

■ 8. Geometry, printing and mass (R9, R13, R14)

- **Runner.** The model's envelope is 200 x 200 x 50 mm, which fits a 220 x 220 x 250 mm bed. Printed mass is about 525 g and print time about 21 h at 25 g/h, inside the 24 h target but only just. A 0.6 mm nozzle would shorten it.
- **Setting height.** The model puts the nozzle centerline 280 mm above normal tailwater (TRL 2: 440 mm) by shortening the frame legs to 80 mm. The runner underside is 210 mm above tailwater, which is how far the tailwater can rise before it touches the runner. At the design point the setting height is 12.3 % of the total site drop (TRL 2: about 18 %).
- **Mass.** Table 6 gives the mass of the turbine unit.

Table 6. Mass estimate.

Item	Mass
Generator, 500 W low-speed BLDC (catalog class, assumed)	7.00 kg
Housing, 315 mm PVC	3.12 kg
Lid, 12 mm HDPE	1.82 kg
Frame, about 2.0 m of 40 x 40 x 4 mm angle	4.84 kg
Bearing housing, posts, plate and guard (aluminium, model volume)	3.81 kg
Shaft, 316 stainless	0.75 kg
Runner, PETG	0.53 kg
Coupling	0.30 kg
Turbine unit, items 1 to 6 and 11	22.2 kg
Manifold and nozzles (about 2.0 m of 90 mm PVC and fittings)	3.20 kg
Gate valve	2.00 kg
All of the above	27.4 kg

R14 is met with the turbine unit taken as items 1 to 6 and 11, the same basis as the TRL 2 estimate of about 20 kg. Amish confirmed this definition on 2026-09-25 (PCF-DDR-002): the manifold and valve are pipework that is carried and fitted separately.

■ 9. Penstock surge (R17)

The pressure wave travels at 301 m/s in 125 mm SN8 PVC, so any closure faster than 0.13 s acts as instantaneous. An instantaneous stop at 3.0 m head (1.10 m/s in the pipe) would add 33.8 m of head, far beyond drainage pipe. Closing the gate valve over 10 s adds only 0.45 m, so the peak at the valve is 33.3 kPa (static 28.9 kPa), within the assumed 50 kPa. A multi-turn gate valve needs about ten turns to close, which enforces the 10 s. A single nozzle blocking at once would add about 16.9 m. The 6 mm screen keeps out anything that could plug a 20 mm or larger insert in one piece, but a mat of leaves could not be ruled out, so the forebay screen must be kept clear.

■ 10. Dump load (R7)

The largest output on the 12 V side is 178.8 W (3.0 m, 15 L/s), so the 300 W dump load has a margin of 1.68 times. The control behavior in R7 (speed within 10 %, full diversion within 1 s) depends on the controller firmware and needs a bench test at TRL 4.

■ 11. Cost (R15)

From `bom/bom.csv`, the kit (every item except the penstock and battery) costs \$448.00 with a new generator, \$2.00 under the \$450 budget. With a salvaged washing machine motor at about \$40 it costs \$378.00. The penstock (\$140.00 for 20 m of 125 mm pipe, up from \$100.00 for 110 mm) and the battery (\$160.00) are excluded, so the penstock change does not affect R15. Amish accepted the \$2 margin on 2026-09-25 (PCF-DDR-002); prices are to be firmed with quotes before any build, which is TRL 4 work and on hold.

Note on the inlet valve: `bom/bom.csv` lists a 90 mm (3 in) gate valve, while this note treats the valve as full bore on the penstock (loss coefficient 0.15 on the penstock velocity head). A 90 mm valve on a 125 mm line needs a reducer and an expander, which would add loss and could erode the 2.8 W margin on R3. This is raised as a new item in `docs/REVIEW.md` ; it is not resolved here.

■ 12. Numbers corrected from TRL 2 and from v0.1

Quantity	TRL 2 estimate	v0.1 (110 mm penstock)	v0.2 (125 mm penstock)
Pipe loss	10 %	23.6 %	17.0 %
Jet diameter	about 33 mm	34 mm	33 mm
Best speed	about 340 rpm	311 rpm	324 rpm
Into the battery at 2.0 m	about 90 W	77.2 W	82.8 W
Water to wire	about 46 %	39.5 %	43.2 %
Daily energy	about 2.2 kWh	1.85 kWh	1.99 kWh
Output at 1.0 m / 3.0 m	about 30 W / 165 W	23.9 W / 146.4 W	26.0 W / 156.4 W
Runaway open circuit at 3.0 m	75 to 83 V	76.4 V	79.5 V (bus clamped at 48 V)
Nozzle height above tailwater	0.44 m	0.28 m	0.28 m
Turbine unit mass	about 20 kg	22.2 kg	22.2 kg
Kit cost, new generator	about \$424	\$448.00	\$448.00